

1 **Title:**

2 VisionMD-Gait: Scalable Clinical Gait Assessment from Smartphone Videos

3 **Authors**

4 Shuyu Liu,¹ Alvin Wong,² Si Chen, MD,³ Patrick J. Antonelli, MD,³ and Diego L. Guarín, Ph.D.^{1,4,5}

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6 ¹ Department of Applied Physiology and Kinesiology, University of Florida, Gainesville, FL

7 ² Department of Computer & Information Science & Engineering, University of Florida, Gainesville, FL

8 ³ Department of Otolaryngology – Head & Neck Surgery, College of Medicine, University of Florida, Gainesville,
9 FL

10 ⁴ J. Crayton Pruitt Family Department of Biomedical Engineering, University of Florida, Gainesville, FL.

11 ⁵ Fixel Institute for Neurological Disease, College of Medicine, University of Florida, Gainesville, FL

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17 Correspondence:

18 Diego L. Guarín, Ph.D.

19 Department of Applied Physiology and Kinesiology, University of Florida, Gainesville, FL

20 Florida Gymnasium, 1864 STADIUM RD GAINESVILLE, FL 32611

21 Email: d.guarinlopez@ufl.edu

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Abstract

VisionMD-Gait enables clinical-grade gait assessment using a single frontal-view smartphone video. This open-source platform integrates state-of-the-art monocular video analysis and 3D pose estimation to compute objective gait parameters without specialized hardware, technical expertise, or sharing data with external services. We validated VisionMD-Gait against a research-grade wearable system in 24 healthy adults and 10 individuals with vestibular dizziness. Video-derived measures showed strong agreement with wearable sensors across gait speed ($r = 0.89$), cadence ($r = 1.00$), and other clinically relevant gait measurements, with mean absolute errors under 8%. Clinical comparisons revealed significant gait alterations in patients with dizziness, despite no clinically overt gait impairments. VisionMD-Gait's ability to process data locally, preserve patient privacy, and function in standard clinical spaces underscores its scalability and transformative potential for gait screening, fall risk assessment, and remote monitoring. VisionMD-Gait represents a step towards the democratization of quantitative gait analysis for clinicians and researchers seeking accessible, cost-effective, mobile health solutions.

1. Introduction

Accurate assessment of gait is pivotal for diagnosing and managing a wide range of neurological, musculoskeletal, and vestibular disorders, including Parkinson's disease, cognitive impairments, spinal cord injury, stroke, and dizziness.^{1–6} Gait impairments, such as reduced stride length, asymmetry, and prolonged double support time, can serve as indicators of pathology or markers of increased fall risk and mobility decline. Yet despite its clinical value, gait assessment remains underutilized in routine clinical care and when performed, are typically limited to subjective observation and the use of standardized rating scales. However, the above approaches require specialized training and may lack the sensitivity needed to detect subtle or progressive changes associated with disease progression or therapeutic intervention.

Digital health technologies, including motion capture systems, wearable sensors, and multi-camera markerless systems, have facilitated objective gait analysis and enable precise quantification of spatiotemporal gait parameters that can help detect and manage gait and mobility impairments.⁷ However, these systems are often expensive, require specialized devices, controlled environments, and technical expertise, making them impractical for widespread use in primary care, community-based, or home-based settings.

Recent advances in computer vision and machine learning have enabled video-based gait analysis using a single camera, offering a potentially scalable solution for gait analysis.^{6,8–11} However, most existing methods rely on sagittal-view recordings, requiring open, unobstructed spaces, limiting their use in clinical environments. Moreover, few single camera methods have been validated against ground truth measures, across diverse patient populations, or in real-world settings.

In this study, we introduce VisionMD-Gait, a scalable, user-friendly, and open-source platform designed to estimate spatiotemporal gait parameters from frontal-view videos.¹² Different from other tools for video-based gait analysis, VisionMD-Gait is a code-free platform, requiring minimal training and expertise; furthermore, VisionMD-Gait processes data locally, maximizing patient privacy and data protection by ensuring that the clinical data does not leave the computer where it is stored.

VisionMD-Gait leverages recent advances in 3D pose estimation and video-based reconstruction of gait kinematics to compute clinically relevant gait parameters from frontal-view smartphone recorded videos. Figure 1 shows an overview of the VisionMD-Gait pipeline which includes three primary components: (i) Subject Selection, (ii) Segment Selection, and (iii) Analysis. In the Subject Selection component, users upload a gait video to VisionMD-Gait and the software automatically localizes the position of the subject of interest in the video. In the Segment Selection component, users define the start and end times of the different gait segments in the video using frame-level precision. Finally, in the Analysis component, VisionMD-Gait uses machine learning algorithms for 3D pose and gait kinematics reconstruction to track body movements during gait and compute quantitative gait parameters that help characterize any potential gait impairments.

VisionMD-Gait was designed for use in unconstrained clinical settings, requiring only a smartphone camera. We validated VisionMD-Gait against a research-grade wearable motion capture system in healthy adults and individuals with dizziness. Results demonstrate strong correlation between video-derived and sensor-derived gait parameters and show that VisionMD-Gait can detect significant differences in gait parameters between healthy controls and people with dizziness.

VisionMD-Gait is an open-source tool that represents an accessible, scalable solution for gait assessment in clinical and research settings and complements subjective observations and standardized rating scales. The software tool is available is freely available for Windows, MacOs, and Linux (www.VisionMD.ai). This work contributes to the growing field of digital health technologies for mobility and demonstrates the feasibility of low-cost, high-impact tools for routine gait monitoring.

2. Results

2.1 Validation

To validate the output of VisionMD-Gait, we performed a validation study comparing video-derived gait parameters provided by VisionMD-Gait against a research grade wearable 3D motion capture system (Noraxon Ultium Motion) in a diverse population that included 24 healthy adults (13 females, 11 males; age: 51.96 ± 24.23 years) and 10 adults with dizziness receiving ambulatory clinical care at a vestibular clinic (8 females, 2 males, age: 51.73 ± 24.77 years). VisionMD-Gait and the wearable system were used to extract gait parameters, including gait speed, cadence, stance time, swing time, double support time, step time, and step length, from gait events corresponding to steady-state gait in an 8-meter walkway. Subjects were instructed to walk back and forth along the walkway, resulting in multiple gait events per subject. Gait parameters analyzed in this study correspond to the average obtained for each gait event, that is, we did not analyze the values obtained from each individual step.

The gait parameters extracted using VisionMD-Gait demonstrated excellent agreement with the reference wearable sensor system. Figure 2 shows scatter plots comparing video- and sensor-based measures across all gait events analyzed in this study (290 total gait events). The figure summarizes bilateral averages for the left and right legs. Table 1 reports the mean $\pm$ SD values for each gait parameter derived from the videos and wearable sensors, along with Pearson correlation coefficients (r) and mean absolute error percentage (MAE%). Video-based measures correlated strongly with sensor-derived measures across spatiotemporal, temporal, and spatial parameters. Gait speed, cadence, step duration, and stance time showed excellent agreement ($r = 0.89, 1.00, 1.00$, and 0.97 , respectively). Swing time, double support time, and step length also demonstrated robust correlations ($r = 0.86, 0.75$, and 0.84 , respectively). The error between video and sensor-based measures was minimal, all measures provided a mean absolute error lower than 8% of the sensors-based measure.

2.2 Clinical Application

To evaluate the clinical utility of VisionMD-Gait, we compared the video-derived gait parameters between healthy controls and persons receiving clinical care due to dizziness. Comparisons were performed using a Linear Mixed Model (LMM) that included age and gender as fixed effects. Random intercepts were used to account for subject-specific variability and the presence of multiple measurements per participant.

Table 2 compares the gait parameters between healthy controls and individuals with dizziness, revealing that there were significant differences across all measures, underscoring clinically meaningful alterations in gait despite the absence of clinically overt gait impairments. The dizziness group exhibited reduced gait speed (1.30 ± 0.17 m/s vs. 1.06 ± 0.12 m/s, $p = 0.001$) and cadence (108.45 ± 6.70 steps/min vs. 100.97 ± 7.27 steps/min, $p = 0.001$), alongside longer step duration, stance time, swing time, and double support time ($p = 0.001$ in all cases). Finally, step length was slightly decreased in the dizziness group (0.72 ± 0.08 m vs. 0.63 ± 0.06 m, $p = 0.029$).

3. Discussion

Falls are a leading cause of mobility loss, hospitalization, and reduced quality of life, particularly in older adults. Quantitative gait assessment is a critical tool for identifying individuals at heightened risk of falls.¹³ For example, decreased gait speed, prolonged stance phase and double support duration, and reduced cadence are well-established markers of increased fall risk.^{14–18} However, obtaining such gait measures typically requires costly sensors, controlled environments, and highly trained personnel. As a result, quantitative gait assessments remain largely confined to research settings or specialized rehabilitation clinics, severely limiting their accessibility for routine clinical care and large-scale screening in community or primary care settings.

VisionMD-Gait addresses these critical barriers by providing a scalable, accessible, and user-friendly platform for clinical-grade spatiotemporal gait analysis using only frontal-view videos recorded with a smartphone in unconstrained settings. In our validation study, we demonstrate high agreement between video-derived measures and gold-standard wearable sensor data across multiple gait parameters, including gait speed ($r = 0.89$), cadence ($r = 1.00$), step duration ($r = 1.00$), stance time ($r = 0.97$), swing time ($r = 0.86$), double support time ($r = 0.75$), and step length ($r = 0.84$), with mean absolute errors lower than 8% of the sensor-based measures for all measures. These results confirm that VisionMD-Gait provides accurate, robust, and clinically relevant gait metrics that agree with traditional sensor-based approaches.

Furthermore, our clinical analysis demonstrated that VisionMD-Gait can identify subtle but significant gait alterations in individuals with dizziness when compared to healthy controls, including slower gait speed, reduced cadence, increased step duration, prolonged stance phase and double support time, and reduced step length. These findings are consistent with a cautious gait strategy, characterized by deliberate, slower limb movements. Such adaptations have been documented in vestibular-compromised individuals and are recognized predictors of reduced mobility and elevated fall risk.^{3,15–17,19} Importantly, this demonstrates VisionMD-Gait's potential to uncover subclinical gait impairments in individuals who do not report overt gait complaints but may still be at elevated risk for falls. Such early identification could be transformative for prevention efforts and to evaluate the effectiveness of interventions aimed at improving gait and balance in both clinical and community settings.

Our results align with a growing body of literature validating monocular, frontal-view video approaches for gait analysis. Azhand et al. (2021) reported high agreement between algorithmically extracted gait measures from frontal-view videos and an instrumented walkway.²⁰ Barzyk et al. (2024) validated a smartphone camera approach in stroke patients, achieving spatiotemporal metrics comparable to those provided by a Motion Capture System.²¹ Stenum et al. (2024) demonstrated that video-based systems can capture clinically relevant, condition-specific gait parameters and track within-participant changes over time.⁹ Similarly, Cimorelli et al. (2024) introduced a portable, in-clinic system for video-based quantitative gait assessments, demonstrating high accuracy in prosthetic users compared with wearable sensors.¹⁰

VisionMD-Gait builds on these advances but introduces key innovations. Unlike most existing methods that require technical expertise for setup and data analysis, VisionMD-Gait provides an end-to-end solution that does not require any programming or specialized knowledge and processes all data locally without using any external servers or cloud processing solutions. This eliminates the need to share sensitive patient videos with third parties, addressing critical privacy and data security concerns. Moreover, VisionMD-Gait is platform-agnostic and can run on standard computers, making it highly scalable and adaptable for use in diverse settings, allowing clinicians, nurses, technicians, and even patients themselves to collect and analyze gait data without specialized training or infrastructure.

Despite its promise, VisionMD-Gait has some limitations. First, while the algorithm was robust across participants, slight inaccuracies may occur in cases of poor lighting or occlusions. Future efforts will focus on refining the algorithm for challenging recording conditions and expanding validation to include diverse clinical populations. Additionally, our current analysis excluded turning events to focus on straight walking segments. Incorporating turning metrics might provide additional clinical information and is an important next step to further enhance VisionMD-Gait clinical utility.²²

In summary, VisionMD-Gait represents a transformative step toward democratizing access to quantitative gait assessments. This tool enables clinicians and researchers to obtain accurate, objective gait parameters without the need for expensive hardware or technical expertise. VisionMD-Gait has the potential to facilitate earlier identification of mobility impairments, personalize rehabilitation strategies, and track disease progression from smartphone videos, which can be easily captured in the clinic or at home environments. Its ability to detect subtle gait differences in patients with dizziness highlights its promise for screening and monitoring. The integration of

VisionMD-Gait into routine practice could profoundly impact the early detection and management of gait abnormalities, fall prevention strategies, and research on mobility disorders and aging.

Methods

Participants

This study recruited 24 healthy participants and 10 patients with vestibular dizziness. All participants were recruited through the vestibular clinic at UF Health ENT and Allergy – The Oaks. Participants in the dizziness group were either diagnosed with benign paroxysmal positional vertigo (BPPV) or reported symptoms consistent with vestibular dizziness. Healthy controls were recruited from relatives of patients and from other individuals seen at the same clinic without vestibular complaints. Participants in the dizziness cohort were excluded if their dizziness or imbalance was attributable to orthostatic hypotension, stroke, Parkinson's disease, cerebral atrophy, paralysis, or other neurological or neuromuscular disorders. Healthy control participants were included only if they had no history of neurological disorders, did not require the use of a cane or armchair for support, and were cognitively intact.

Procedure

Gait events were recorded using a smartphone (iPhone 12) at 60 frames per second (fps) with a resolution of 1080 × 1920 pixels. To validate the markerless method, a wearable 3D motion capture system, Noraxon Ultium Motion™ IMUs (Noraxon, Inc., Scottsdale, AZ), was used simultaneously to collect kinematic gait data. This system employs multiple inertial measurement units (IMUs) placed on various body segments to track acceleration, movement, and three-dimensional (3D) position, enabling detailed analysis of gait kinematics.

Figure 3 A) illustrates the wearable sensors arrangement; participants wore sensors placed bilaterally on their feet, shanks, thighs, and a single sensor on the lower back to capture comprehensive gait metrics. Figure 3 B) presents a schematic of the data collection setup, depicting an 8-meter walkway with designated start and end points and the position of the smartphone camera used for video data acquisition. Participants were instructed to walk at a comfortable, self-selected speed, turn upon reaching the end of the walkway, and complete at least four laps, depending on their health status and comfort level. Videos and wearable sensor data were acquired concurrently and later manually segmented, directly within VisionMD-Gait, to exclude turns, ensuring that the estimated gait parameters reflected steady-state walking. Figure 3 C) shows the real-world recording setup employed in this study. All data were acquired in a vestibular clinic hallway (UF Health The Oaks, Gainesville, FL) using a smartphone supported on a tripod, positioned behind the participant at a distance sufficient to capture their full body. The video-based workflow required no adjustments to the clinic environment or patient routines, underscoring the practical clinical utility and scalability of VisionMD-Gait for real-world implementation. In contrast, the sensor-based workflow required careful positioning of the sensors and an additional sensors calibration step, often resulting in missed trials that needed to be repeated due to sensors movement.

All gait videos were analyzed using the VisionMD-Gait processing pipeline; Figure 4 illustrates the full processing workflow of VisionMD-Gait which consists of three main components: a pose estimation model, a gait transformer model, and a feature extractor. Pose estimation was performed using MeTRAbs, a monocular 3D human pose estimator model trained on 28 open-sourced datasets with high-quality 3D human pose labels.²³ MeTRAbs receives as input the video frames along with a bounding box indicating the position of the subject in the frame, which was estimated using a YOLO-based object detector trained to identify human bodies. The model's output is the 3D position of multiple joint coordinates in metrics units with the position of the camera as the origin.

The resulting joint coordinates were re-ordered to align with the Human3.6M Dataset format and were re-centered at the pelvis. These standardized joint sequences, along with the participant's height, were passed to the gait transformer model, which estimated lower-limb joint kinematics (hip, knee, and ankle angular displacement and velocities) and detected foot contact events, including heel strikes and toe-offs.²⁴ Gait phase

parameters, including stance time, swing time, and double support time, were computed based on the timing of these gait events. Step length, gait speed, and cadence were derived from changes in the estimated distance of the pelvis to the camera at each video frame provided by MeTRAbs. Table 3 summarizes the special, temporal, and spatiotemporal gait parameters derived from videos processed with VisionMD-Gait.

To mitigate potential asymmetrical biases between the left and right sides, we mirrored each video and repeated the entire processing pipeline. The final gait parameters values were averaged across the results obtained from the original video and mirrored versions.

Ground truth kinematic gait parameters were obtained from the wearable sensors using the MR3.20 software (Noraxon, Inc., Scottsdale, AZ). To validate the video-based analysis against sensor-based analysis, we computed the Pearson correlation coefficient between the two methods. Additionally, as shown in equation 1, we calculated the mean absolute error (MAE%) as a percentage of the video-based measure to assess the accuracy of the video-based estimates.

$$MAE\% = \left(\frac{ABS[SensorMeasure - VideoMeasure]}{SensorMeasure} \right) \times 100\% \quad (1)$$

To compare kinematic gait differences between participants with vestibular dizziness and healthy controls, we employed a Linear Mixed Model (LMM). This approach facilitates the analysis of repeated measures while accounting for both fixed and random effects. Age and gender were included as covariates to control for demographic variability. Given the presence of multiple walking trials per participant, random intercepts were added to the model to account for individual baseline differences. Equation 2 describes the structure of the model, where j denotes each participant, i represents repeated trials within participant j . y_{ij} is the gait measure (e.g., step time, step length) for the i -th trial of the j -th participant. $Group_{ij}$ is a binary indicator (0 = healthy control, 1 = vestibular dizziness). u_{0j} is the random intercept for subject j , and ϵ_{ij} is the residual error for each trial.

$$y_{ij} = \beta_0 + \beta_1 Group_{ij} + \beta_2 Age_{ij} + \beta_3 Gender_{ij} + u_{0j} + \epsilon_{ij} \quad (2)$$

Each β coefficient has an associated p-value, which tests the null hypothesis that the coefficient is equal to zero (no effect). For this study, we considered p-values less than 0.05 as statistically significant, indicating the observed effect is unlikely to have occurred by chance. We implemented all models in Python using the *statsmodels* package and fitted them using restricted maximum likelihood estimation with the L-BFGS optimizer.

Data availability

The datasets used and/or analyzed during the current study are available at <https://github.com/mea-lab/GaitValidation>

Code availability

The underlying code for this study is available at <https://github.com/mea-lab/GaitValidation>

VisionMD, the open-source software used to process all the videos can be downloaded free of charge from www.visionmd.ai

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Author contributions

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SL: Writing, review & editing, Visualization, Formal analysis, Data acquisition, Data curation. AW: Software development, Methodology. SC: Writing – review & editing, Methodology, Data acquisition, Conceptualization. PJA: Writing – review & editing, Methodology, Data acquisition, Conceptualization. DLG: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Ethics Declaration

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Competing interests

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The authors declare no competing interests.

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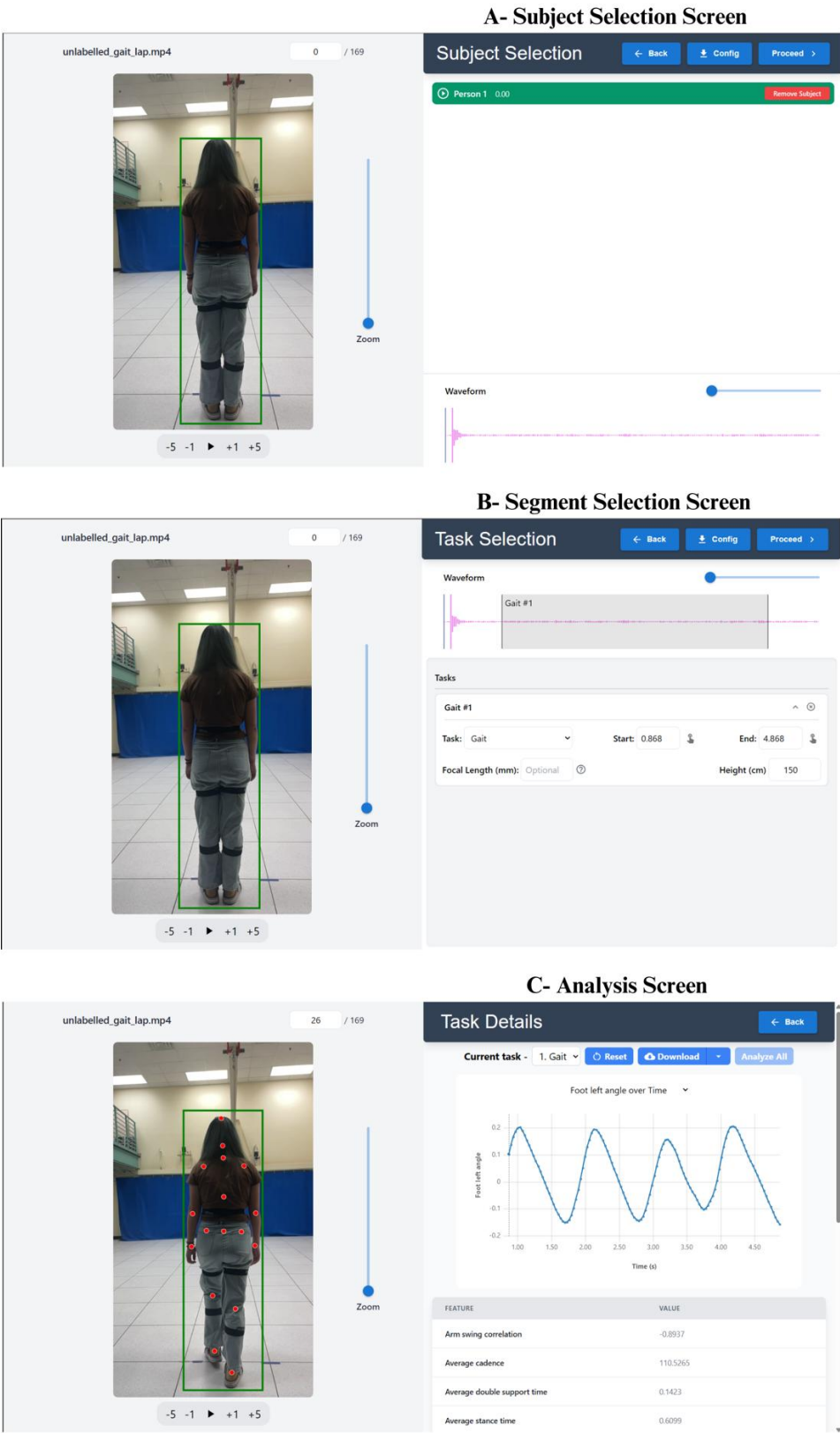
Human Ethics and Consent to Participate declarations

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The study was approved by the University of Florida's Institutional Review Board, ethical approval number: IRB202302162. This study was conducted in accordance with the principles of the Declaration of Helsinki, all participants provided written informed consent prior to enrollment.



Step 1: A video is uploaded into VisionMD-Gait.

Step 2: Video is automatically processes to localize the persons in the video.

Step 3: If more than one person is in the video, the user must selects the subject to be analyzed.

Step 4: The user localizes the start and end of each gait event in the video using the waveform display.

Step 5: Each gait event receives an independent name. Additional information, such as the camera parameters and the subject’s height can be provided.

Step 6: VisionMD-Gait automatically estimates and tracks the position of multiple body landmarks

Step 7: Landmark tracking results are displayed to verify their accuracy.

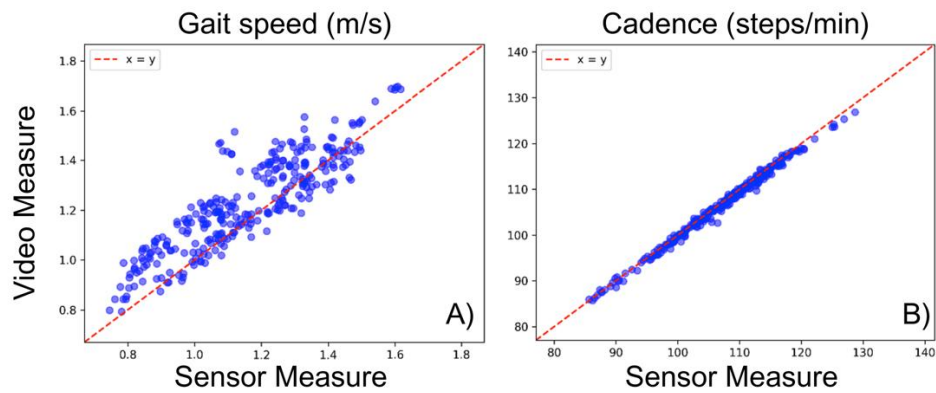
Step 8: VisionMD-Gait estimates joint kinematics (angular displacement and velocity) and gait events from the video.

Step 9: VisionMD-Gait computes and displays multiple gait parameters using the estimated landmarks and joint kinematics.

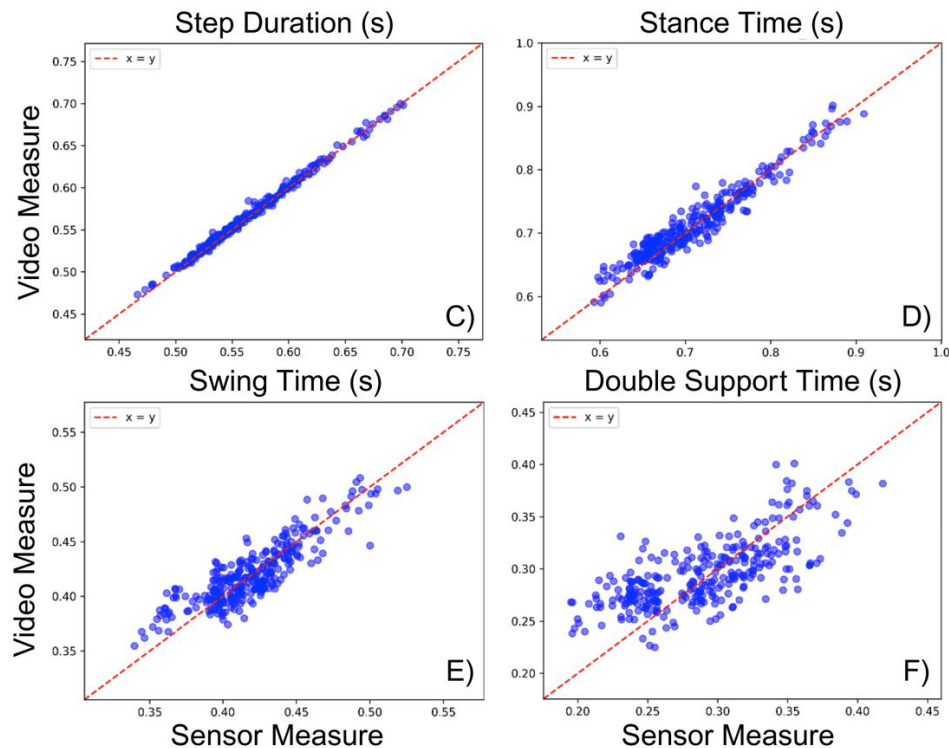
Step 10: Joint kinematics and gait parameters can be stored for future use.

Figure 1. Overview of the VisionMD-Gait processing pipeline. VisionMD-Gait integrates state-of-the-art monocular video analysis methods into an open-source platform for clinical gait assessment. (i) Subject Selection: VisionMD-Gait automatically detects and tracks the subject of interest in a frontal-view video. (ii) Segment Selection: Users define gait segments with frame-level precision. (iii) Analysis: The system reconstructs 3D kinematics and computes spatiotemporal gait parameters including gait speed, step length, cadence, stance time, swing time, and double support time. Kinematics signals and gait parameters can be stored for future use in external tools.

Spaciotemporal Parameters



Temporal Parameters



Spatial Parameters

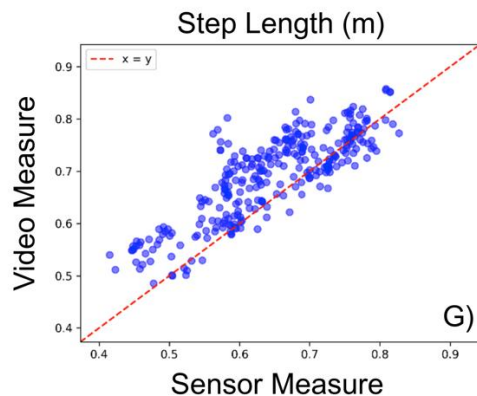


Figure 2. Validation of VisionMD-Gait against wearable sensors. Scatter plots comparing gait parameters derived from monocular smartphone videos using VisionMD-Gait versus wearable sensor data. A) Gait speed, B) Cadence, C) Step duration, D) Stance time, E) Swing time, F) Double support time, and G) Step length. The red dashed line indicates perfect agreement ($x=y$). VisionMD-Gait demonstrated strong agreement across all measures, validating its utility for clinical gait analysis.

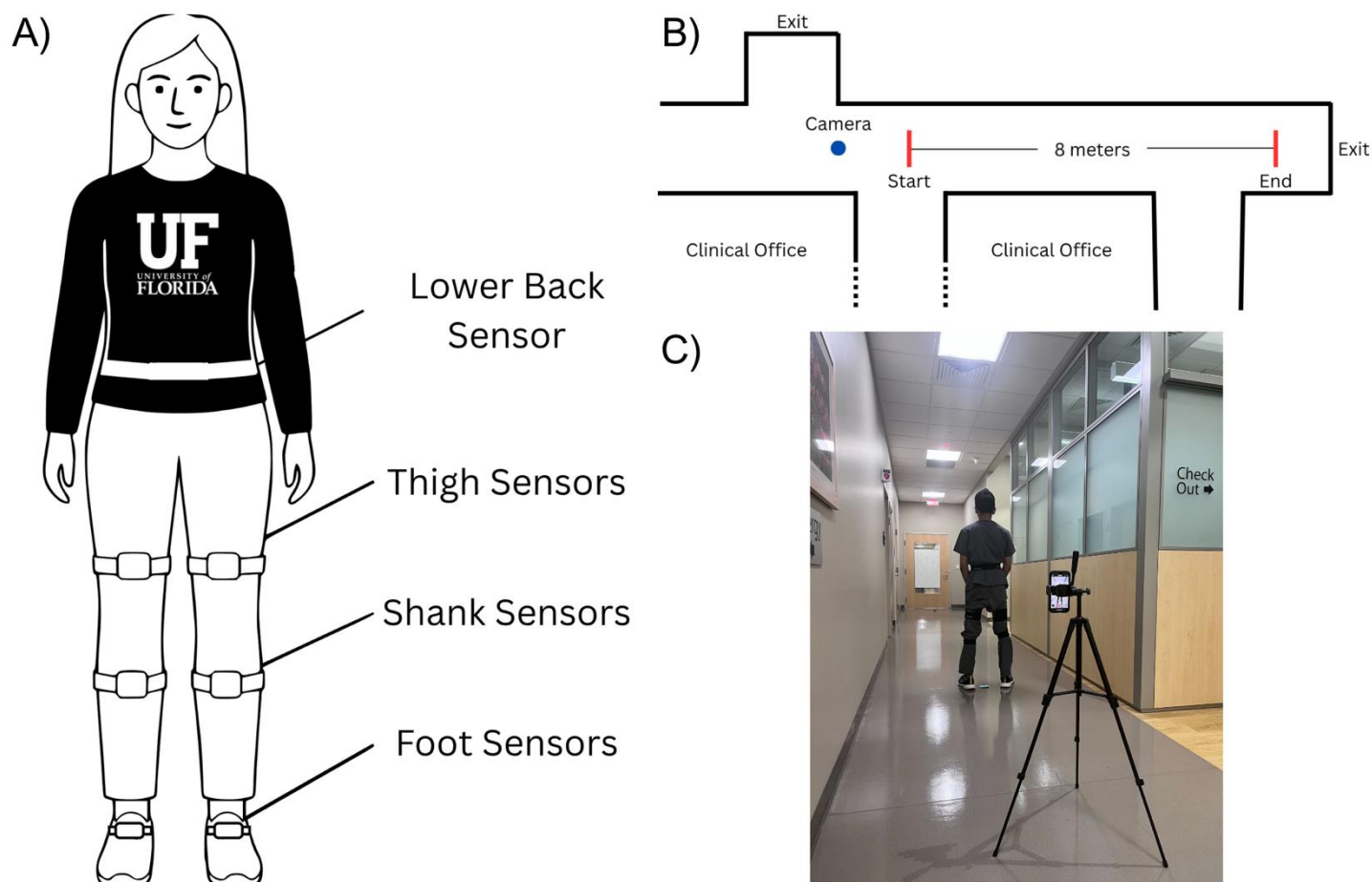


Figure 3. Validation study setup and clinical environment. (A) Placement of wearable inertial sensors on the lower back, thighs, shanks, and feet for estimation of ground truth gait measurements. (B) Schematic of the clinical hallway used for video and sensor data acquisition. Participants walked an 8-meter path while a smartphone camera recorded from a fixed position. (C) Real-world recording setup showing smartphone camera placement and the participant performing the gait task.

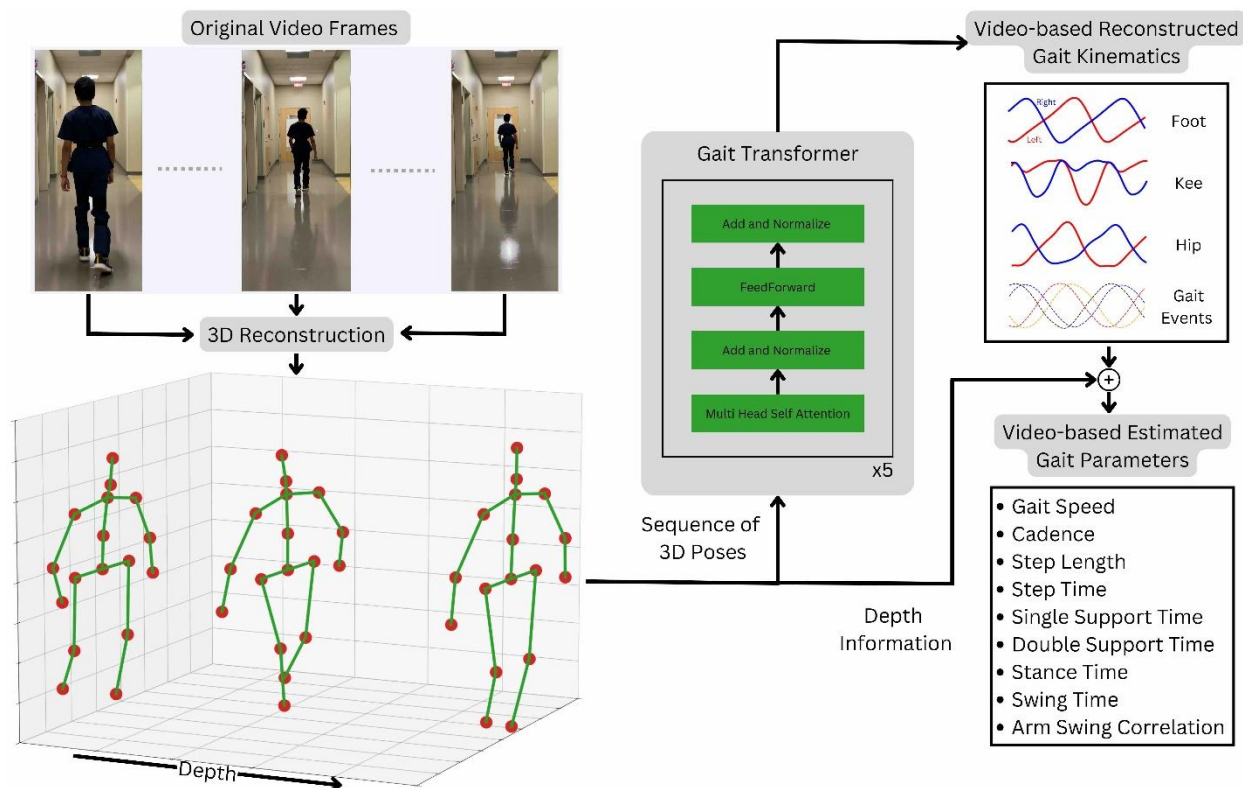


Figure 4. Overview of the VisionMD-Gait processing pipeline. Original video frames are processed to reconstruct 3D human poses, which are then passed through a gait transformer model to extract joint kinematics and gait events. From these outputs, spatiotemporal gait parameters such as gait speed, cadence, step length, stance time, and swing time are estimated.

Tables

Table 1. Agreement between video-based and sensor-based gait parameters. Mean values $\pm$ standard deviation of gait parameters measured using VisionMD-Gait and wearable sensors. The mean absolute error (MAE) and Pearson correlation coefficient (r) quantify accuracy and agreement between the two methods.

Gait Parameter	Sensor measure	Video Measure	Mean Absolute Error (%)	Pearson Correlation (r)
Gait Speed (m/s)	1.16 $\pm$ 0.20	1.23 $\pm$ 0.19	6.03	0.89
Cadence (steps/min)	106.68 $\pm$ 7.80	106.21 $\pm$ 7.67	0.44	1.00
Step Duration (s)	0.57 $\pm$ 0.04	0.57 $\pm$ 0.04	0.31	1.00
Stance Time (s)	0.71 $\pm$ 0.06	0.72 $\pm$ 0.06	1.41	0.97
Swing Time (s)	0.42 $\pm$ 0.03	0.42 $\pm$ 0.03	0.24	0.86
Double Support Time (s)	0.29 $\pm$ 0.04	0.30 $\pm$ 0.03	3.44	0.75
Step Length (m)	0.65 $\pm$ 0.10	0.70 $\pm$ 0.08	7.69	0.84

Table 2. Comparison of video-derived gait parameters between healthy controls and individuals with dizziness. Values represent mean ± standard deviation. p-values indicate the significance of the group coefficient in the linear mixed-effects model, adjusted for age, gender, and repeated measures per participant.

Gait Parameter	Healthy Control Group	Dizziness Group	p-value
Gait Speed (m/s)	1.30 ± 0.17	1.06 ± 0.12	0.001
Cadence (steps/min)	108.45 ± 6.70	100.97 ± 7.27	0.001
Step Duration (s)	0.55 ± 0.04	0.60 ± 0.04	0.001
Stance Time (s)	0.70 ± 0.04	0.76 ± 0.06	0.001
Swing Time (s)	0.41 ± 0.02	0.44 ± 0.03	0.001
Double Support Time (s)	0.29 ± 0.04	0.32 ± 0.03	0.001
Step Length (m)	0.72 ± 0.08	0.63 ± 0.06	0.029

Table 3. Description of gait parameter computations. Gait features were computed using heel strike, toe-off timestamps, and depth data from pose estimation. Metrics follow standard definitions to enable consistent comparison across methods.

Gait Parameter	Definition	Computation / Formula
Temporal parameter		
Left Swing Time	Time the left foot is in the air	Timestamp (Left Heel Strike) – Timestamp (Previous Left Toe Off)
Right Swing Time	Time the right foot is in the air	Timestamp (Right Heel Strike) – Timestamp (Previous Right Toe Off)
Left Stance Time	Time the left foot is on the ground	Timestamp (Left Toe Off) – Timestamp (Previous Left Heel Strike)
Right Stance Time	Time the right foot is on the ground	Timestamp (Right Toe Off) – Timestamp (Previous Right Heel Strike)
Double Support Time	Time both feet are in contact with the ground	Timestamp (Left Toe Off – Previous Right Heel Strike) + Timestamp (Right Toe Off – Previous Left Heel Strike)
Left Step Time	Time between consecutive heel strikes when the first heel strike is left	Timestamp (Left Heel Strike) – Timestamp (Previous Right Heel Strike)
Right Step Time	Time between consecutive heel strikes when the first heel strike is right	Timestamp (Right Heel Strike) – Timestamp (Previous Left Heel Strike)
Average Step Time	Mean of left and right step times	(Left Step Time + Right Step Time) / 2
Spatial parameter		
Left Step Length	Distance between heel strikes along depth axis	Depth (Left Heel Strike) – Depth (Right Heel Strike)
Right Step Length	Distance between heel strikes along depth axis	Depth (Right Heel Strike) – Depth (Left Heel Strike)
Spaciotemporal parameter		
Gait Velocity	Average walking speed	(Step Length at Last Right Heel Strike – Step Length at First Right Heel Strike) / (Timestamp at Last Right Heel Strike – Timestamp at First Right Heel Strike)
Cadence	Steps per minute	(1 / Average Step Time) × 60

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